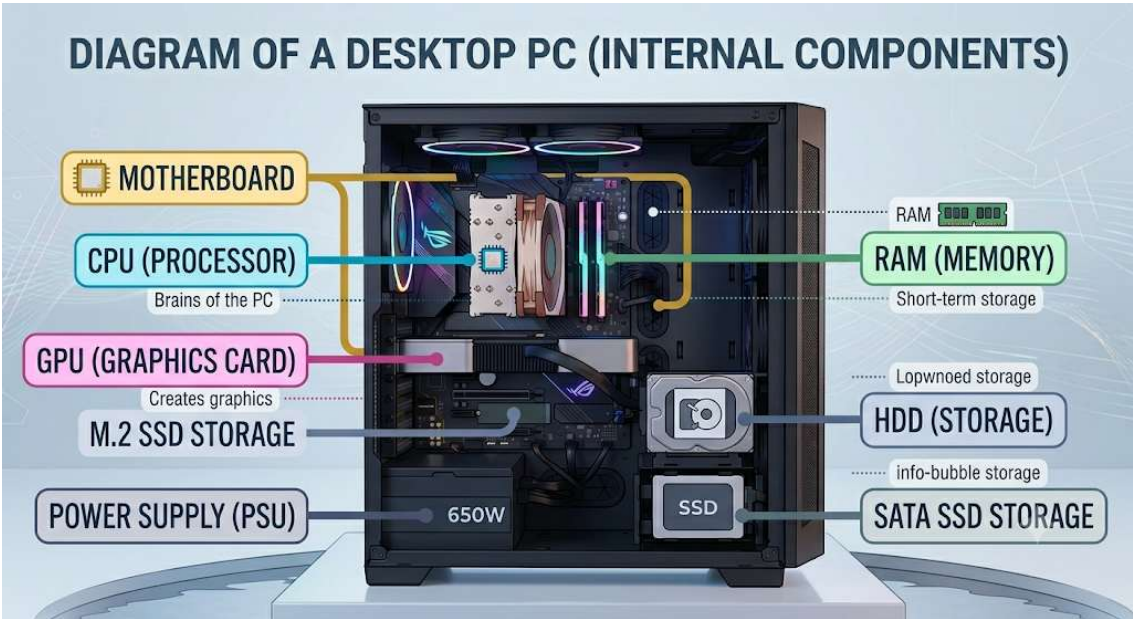


1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Comparison Table

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	 Slower	 Super Fast

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
	Takes more time to boot up your computer and load large files.	Boots up your PC in seconds and loads apps/games almost instantly.
Durability	 Fragile Because it has moving parts, dropping it can permanently damage your data.	 Highly Shock-Resistant No moving parts inside, so it easily survives accidental drops and bumps.
Noise	 Clicky / Whirring You can often hear it spinning or making slight clicking noises when working.	 Completely Silent Runs 100% silently because it is entirely electronic.
Cost & Capacity	 Budget-Friendly Gives you a massive amount of storage space for a very low price.	 More Expensive Costlier per gigabyte, though prices have come down a lot.
Typical Use Cases	 Mass Storage & Backups	 Daily Driving & Speed Essential for modern laptops, gaming PCs, and the drive where

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
	Perfect for desktop PCs, external backup drives, and storing huge movie or photo collections.	your Operating System (Windows/Mac) is installed.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

The Brain: CPU (Central Processing Unit)

The CPU is the director of the game, handling all the background logic and math. It calculates character movements, manages enemy AI, and processes your controller inputs in real-time. Without a fast CPU, your game would constantly stutter as it struggles to keep up with the action.

The Short-Term Memory: RAM (Random Access Memory)

RAM acts as the super-fast temporary storage that holds the map data, weapon models, and player skins while you are actively playing. It allows the game to load assets instantly so you don't face sudden lag while sprinting into a new area. Think of it as a messy but incredibly fast desk where the CPU keeps everything it needs right at arm's reach.

The Artist: GPU (Graphics Processing Unit)

The GPU is responsible for bringing the battleground to life by rendering all the visual elements on your screen. It handles the 3D graphics, explosive particle effects, realistic lighting, and smooth frame rates that let you spot enemies far in the distance. Simply put, it takes the raw data from the CPU and turns it into the beautiful, fast-moving image you actually see.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

1. The Graphics Card (GPU) – The Star Striker

The graphics card is the single most important component for gaming. It is responsible for rendering the stadium atmosphere, realistic grass textures, dynamic lighting, and life-like player faces.

Duelmasters

- **Why Upgrade:** If your current card stutters during crowded penalty-box action or replays, a GPU upgrade will immediately fix it. It ensures your frame rate stays locked at a smooth 60 FPS or higher, eliminating lag.
- **What to Aim For:** While the game *can* run on older cards, upgrading to something modern like an **NVIDIA RTX 3060 / 4060** or **AMD Radeon RX 6700 XT** will let you crank the graphics settings to "Ultra" at 1080p or 1440p resolution without a sweat.

Duelmasters

2. The Storage (SSD) – The Fast Break

Gone are the days when traditional spinning Hard Drives (HDDs) cut it. Modern sports titles demand fast data streaming.

- **Why Upgrade:** Upgrading to a Solid State Drive (SSD) drastically minimizes menu loading times and gets you into Ultimate Team or Career Mode matches in just a couple of seconds. Crucially, it also stops mid-game micro-stutters caused when the PC tries to load stadium or crowd assets on the fly.

Duelmasters+ 1

- **What to Aim For:** An **M.2 NVMe SSD** with at least 500GB to 1TB of space. The latest versions of the game require around 100GB of free space.

LagoFast Game Booster

3. The Processor (CPU) – The Midfield Playmaker

The CPU handles the "brain" of the game: the complex tactical AI of the 21 other players on the pitch, referee decisions, ball physics, and crowd animations.

- **Why Upgrade:** If you pair a powerful new graphics card with an outdated processor, you will hit a "bottleneck." Your GPU will be waiting around for the CPU to pass it data, leading to sudden, frustrating frame drops.
- **What to Aim For:** A solid mid-range processor like an **Intel Core i5 (12th Gen or newer)** or an **AMD Ryzen 5 (5000 series or newer)**. These have plenty of cores to easily manage both the game and background apps like Discord or Spotify.

Duelmasters

4. System Memory (RAM) – The Bench Depth

RAM acts as the short-term memory your PC needs to juggle the active game files quickly.

- **Why Upgrade:** While the absolute minimum requirement is 8GB, running modern titles on 8GB causes Windows to constantly juggle memory, leading to hitching. Upgrading your RAM provides breathing room for smooth multitasking.
 - **What to Aim For:** Upgrade to **16GB of DDR4 or DDR5 RAM**. It is an affordable upgrade that ensures your system runs comfortably without breaking a sweat.
- Duelmasters

Quick Upgrade Checklist

Component	Target Spec for Smooth Gameplay	Impact Level
GPU	NVIDIA RTX 3060 / AMD RX 6600 XT or higher	★★★★★ (Massive visual & FPS boost)
Storage	1TB NVMe M.2 SSD	★★★★ (Incredibly fast loading times)
CPU	Intel i5-12400 / AMD Ryzen 5 5600 or higher	★★★★ (Stable frames, no stuttering)
RAM	16 GB Dual-Channel	★★★ (Smooth multitasking)